

Lando System Specification Language v2 Grammar

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1 Notation

The grammar for the Lando System Specification Sublanguage is written in the EBNF notation. The main elements of the notation that we utilize are:

- Keyword terminals are represented in **typewriter** font.
- Single-character terminals appear in single quotes, e.g. `'!'` or `'\n'`.
- More complex terminals are represented using standard regular expressions written as `«regexp»`, which may include escaped control characters such as `\n`. Regular expressions may use character classes (with negation). Note that the pattern `«/»` matches the literal character forward slash.
- An optional *thing* is written as `[ thing ]`.
- Zero or more repetitions of *thing* are written as `{ thing }`.
- One or more repetitions of *thing* are written as `{ thing }+`.
- If *S* and *T* are non-terminals or regular expressions, *S* – *T* denotes the set of strings matching *S* but not *T*.
- *EOF* represents end-of-file; it can be thought of as a special terminal that is not consumed, but rather is replicated as often as necessary to match the grammar when parsing.
- Whitespace and line separators are significant throughout. Note that the line separator terminal $\mathcal{N}$ consumes any whitespace immediately following the actual end-of-line character(s). Two consecutive line separators (i.e. matching $\mathcal{N}\mathcal{N}$) are always lexed instead as a single blank line terminal $\mathcal{B}$; this distinction is necessary to resolve certain ambiguities in the grammar for paragraphs. The general line separator $\mathcal{L}$ is the union of $\mathcal{N}$ and $\mathcal{B}$.
- In cases of ambiguity, productions involving repetition behave “greedily,” i.e. they match as much of the input as possible. For example, because the Sentence Body production uses `{ }`, the string “**abc def ghi**” parses as a single sentence containing three words, rather than as three sentences of one word each.

1.1 Context-Free Grammar

<i>landoSource</i>	::=	$\{ \mathcal{L} \} \{ body \} \{ \mathcal{L} \} \{ \mathcal{CL} \} \{ \mathcal{L} \} EOF$	<i>Lando source</i>
<i>specElement</i>	::=	$system \mid subsystem \mid subsystemImport \mid component \mid componentImport$ $\mid events \mid scenarios \mid requirements \mid relation$	<i>Top-level element</i>
<i>system</i>	::=	$\{ \mathcal{CL} \} system \text{ name } [abbrev] [\mathcal{C}] \{ \mathcal{L} \}^+$ $explanation \{ \mathcal{CL} \} [indexing \text{ blockend }]$ $[contains \{ \mathcal{L} \} body \text{ end } [\mathcal{C}] \text{ blockend }]$	<i>System</i>
<i>body</i>	::=	$\{ specElement \}$	<i>Source/(Sub)system body</i>
<i>subsystem</i>	::=	$\{ \mathcal{CL} \} subsystem \text{ name } [abbrev] \{ clientClause \} [\mathcal{C}] \{ \mathcal{L} \}^+$ $explanation \{ \mathcal{CL} \} [indexing \text{ blockend }]$ $[contains \{ \mathcal{L} \} body \text{ end } [\mathcal{C}] \text{ blockend }]$	<i>Subsystem</i>
<i>subsystemImport</i>	::=	$\{ \mathcal{CL} \} import \{ S \}^+ subsystem \text{ name } [abbrev] \{ clientClause \} [\mathcal{C}] \text{ blockend}$	<i>Subsystem import</i>
<i>explanation</i>	::=	$paragraph$	<i>Explanation</i>
<i>abbrev</i>	::=	$\{ S \} ' (\{ S \} nameWord \{ S \}) ' \{ S \}$	<i>Abbreviation</i>
<i>inheritClause</i>	::=	$inherit \text{ qname } \{ \mathcal{L} \} \{ ' , ' \{ \mathcal{L} \} \text{ qname } \{ \mathcal{L} \} \}$	<i>Inherit clause</i>
<i>clientClause</i>	::=	$client \text{ qname } \{ \mathcal{L} \} \{ ' , ' \{ \mathcal{L} \} \text{ qname } \{ \mathcal{L} \} \}$	<i>Client clause</i>
<i>indexing</i>	::=	$indexing \{ S \} \{ \{ \mathcal{L} \}^+ indexEntry \}$	<i>Indexing</i>
<i>indexEntry</i>	::=	$indexValue ' : ' indexValuePart \{ \{ \mathcal{L} \}^+ indexValuePart \}$	<i>Index Entry</i>
<i>indexValuePart</i>	::=	$indexValue \{ \mathcal{L} \} [\mathcal{C}]$	<i>Index Value Part</i>
<i>component</i>	::=	$\{ \mathcal{CL} \} component \text{ name } [abbrev] \{ inheritClause \mid clientClause \} [\mathcal{C}] \{ \mathcal{L} \}^+$ $explanation \{ componentFeature \text{ blockend } \}$	<i>Component definition</i>
<i>componentImport</i>	::=	$\{ \mathcal{CL} \} import \{ S \}^+ component \text{ name } [abbrev] \{ clientClause \} [\mathcal{C}] \text{ blockend}$	<i>Component import</i>
<i>componentFeature</i>	::=	$constraint \mid command \mid query$	<i>Component Feature</i>
<i>constraint</i>	::=	$\{ \mathcal{CL} \} sentBody ' . ' [wordSep] [\mathcal{C}]$	<i>Constraint Feature</i>
<i>query</i>	::=	$\{ \mathcal{CL} \} sentBody ' ? ' [wordSep] [\mathcal{C}]$	<i>Query Feature</i>
<i>command</i>	::=	$\{ \mathcal{CL} \} sentBody ' ! ' [wordSep] [\mathcal{C}]$	<i>Command Feature</i>
<i>events</i>	::=	$\{ \mathcal{CL} \} events \text{ name } [\mathcal{C}] \{ \mathcal{L} \}^+ \{ eventEntry \}$	<i>Events</i>
<i>eventEntry</i>	::=	$\{ \mathcal{CL} \} name [\mathcal{C}] \{ \mathcal{L} \}^+ paragraph$	<i>Event Entry</i>
<i>scenarios</i>	::=	$\{ \mathcal{CL} \} scenarios \text{ name } [\mathcal{C}] \{ \mathcal{L} \}^+ \{ scenarioEntry \}$	<i>Scenarios</i>
<i>scenarioEntry</i>	::=	$\{ \mathcal{CL} \} name [\mathcal{C}] \{ \mathcal{L} \}^+ paragraph$	<i>Scenario Entry</i>
<i>requirements</i>	::=	$\{ \mathcal{CL} \} requirements \text{ name } [\mathcal{C}] \{ \mathcal{L} \}^+ \{ requirementEntry \}$	<i>Requirements</i>
<i>requirementEntry</i>	::=	$\{ \mathcal{CL} \} name [\mathcal{C}] \{ \mathcal{L} \}^+ paragraph$	<i>Requirement Entry</i>
<i>relation</i>	::=	$\{ \mathcal{CL} \} relation \text{ qname } \{ inheritClause \mid clientClause \}^+ [\mathcal{C}] \text{ blockend}$	<i>Relation Declaration</i>
<i>qname</i>	::=	$name \mid qname ' : ' name$	<i>Name</i>
<i>name</i>	::=	$\{ S \} nameWord \{ \{ S \}^+ nameWord \} \{ S \}$	<i>Name facet</i>
<i>nameWord</i>	::=	$\ll [^ \backslash r \backslash n \backslash t () :] * [^ \backslash r \backslash n \backslash t () : , . ! ?] \gg - \ll / . * \gg - nonName$	<i>Name word</i>
<i>nonName</i>	::=	$client \mid inherit$	<i>Keywords</i>
<i>indexValue</i>	::=	$\{ S \} indexValueWord \{ \{ S \}^+ indexValueWord \} \{ S \}$	<i>Index value</i>
<i>indexValueWord</i>	::=	$\ll [^ \backslash r \backslash n \backslash t] * [^ \backslash r \backslash n \backslash t :] \gg - \ll / . * \gg$	<i>Index value word</i>
<i>paragraph</i>	::=	$\{ sentence \}^+ parend$	<i>Paragraph</i>
<i>parend</i>	::=	$[\mathcal{C}] (\mathcal{B} \{ \mathcal{L} \} \mid EOF)$	<i>Paragraph end</i>
<i>sentence</i>	::=	$sentBody [sentTerm] [wordSep]$	<i>Sentence</i>
<i>sentBody</i>	::=	$sentWord \{ wordSep \text{ sentWord } \} [wordSep]$	<i>Sentence Body</i>
<i>wordSep</i>	::=	$\{ S \}^+ \mid \{ S \} [\mathcal{C}] \mathcal{L}$	<i>Word Separator</i>
<i>sentWord</i>	::=	$\ll [^ \backslash r \backslash n \backslash t] * [^ \backslash r \backslash n \backslash t . ! ?] \gg - \ll / . * \gg$	<i>Sentence word</i>
<i>sentTerm</i>	::=	$' . ' \mid ' ! ' \mid ' ? '$	<i>Sentence terminator</i>
<i>blockend</i>	::=	$\{ \mathcal{L} \}^+ \mid EOF$	<i>Block end</i>
<i>S</i>	::=	$\ll [\backslash t] \gg$	<i>Whitespace character</i>
<i>N</i>	::=	$[' \backslash r '] ' \backslash n ' [S] \mid ' \backslash r ' [S] \mid EOF$	<i>Single line separator</i>

$\mathcal{B}$	::=	$\mathcal{NN}$	<i>Blank line</i>
$\mathcal{L}$	::=	$\mathcal{N} \mathcal{B}$	<i>General line separator</i>
$\mathcal{C}$	::=	$\langle\langle // [^\wedge \backslash \mathbf{r} \backslash \mathbf{n}]^* \rangle\rangle$	<i>Comment</i>

2 Context-Sensitive Considerations

The following extra-grammatical rules are applied during or after parsing the concrete grammar above.

- If a *system* or *subsystem* element does not have an explicit body, then it has an implicit body that begins immediately after the element and extends until just before the next **system**, **subsystem** or unmatched **end**.
- *subsystemImport* can only appear in an (explicit or implicit) *system* or *subsystem* body. *component* and *componentImport* can only appear in an (explicit or implicit) *subsystem* body.
- A *sentence* that does not end with a '.', '!', or '?' symbol (possibly followed by white space characters or line separators) induces a warning message.
- Note that because of the “greedy” resolution of ambiguity, this warning can only occur at the end of a paragraph.